POLAR NATURE OF WATER AND SIMILAR MOLECULES 8

Water is very important because of its physical and chemical properties, which result from the structure of its molecules. Water has similarities to ammonia (NH_3) and hydrogen sulfide (H_2S). Water and hydrogen sulfide molecules have 'V-shaped' structures. Ammonia is a triangular pyramid. Figure 1 shows the structures of these molecules.

1 Use figure 1 to help you draw Lewis electron dot structures of water, ammonia and hydrogen sulfide. Remember that Lewis electron dot structures show only outer (valence) electrons.

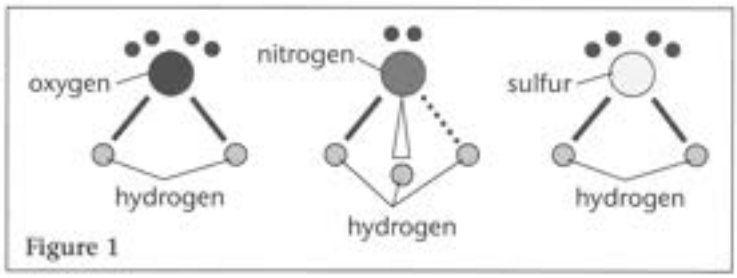


Figure 1

Electrons in a covalent bond can be distributed inequally between two atoms, due to differences in electronegativity (one end is slightly positively charged while the other is slightly negatively charged). When this happens, a polar molecule is formed. Water has polar covalent bonding in its molecules. The part of the molecule that is relatively negative is shown as δ^- and the relatively positive part as δ^+. Figure 2 shows the polarity in a water molecule.

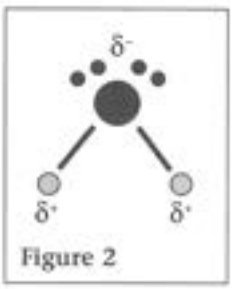


Figure 2

2 Which part of the water molecule is relatively positive?

3 Which part of the water molecule is relatively negative?

The dipoles formed (δ^+ and δ^-) attract dipoles in other molecules. This affects the physical properties of the compound. In water, there is a particularly strong form of dipole-dipole attraction called **hydrogen bonding**.

4 Define *polarity* and *dipole*.

5 Draw a diagram to show hydrogen bonding between two water molecules.

Boiling points of various hydrides

	Compound Group IV	Boiling point (K)	Compound Group V	Boiling point (K)	Compound Group VI	Boiling point (K)	Compound Group VII	Boiling point (K)
Period 2	CH_4	111	NH_3	310	H_2O	373	HF	292
Period 3	SiH_4	161	PH_3	188	H_2S	211	HCl	188
Period 4	GeH_4	185	AsH_3	216	H_2Se	232	HBr	206
Period 5	SnH_4	221	SbH_3	273	H_2Te	271	HI	238

Table 1

6 Use graph paper and refer to table 1.

- **a** Draw your own set of axes as shown in figure 3.
- **b** Plot the boiling point (K) of each hydride from table 1 on the y-axis of your graph.
- **c** Label each hydride.
- **d** Plot Period on the x-axis. (Period 2 is CH_4 to HF, Period 3 is SiH_4 to HCl, etc.)
- **e** Join the boiling points of hydrides in the same Group.

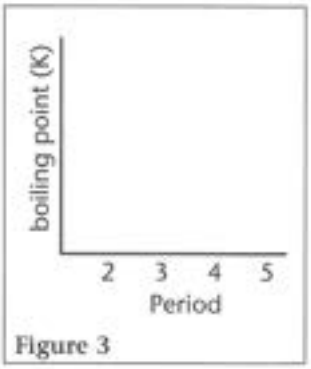


Figure 3

7 Polar molecules have strong intermolecular forces (hydrogen bonds) that result in higher than expected melting and boiling points. Name three compounds from table 1 that you think have hydrogen bonds present.

8 Give the formula for:

- **a** ammonia
- **b** hydrogen sulfide

9 What are *valence* electrons?

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